CHAPTER 11: The Mole

In Science, as previously mentioned, the mass of single atoms is far too small to be of practical value. However, masses must be used in calculations, hence masses based on Atomic (and Formula) Masses are used.

Consider the elements aluminium and carbon —

$$\frac{\text{Atomic mass of aluminium}}{\text{Atomic mass of carbon}} = \frac{27}{12}$$

$$\text{i.e.} \quad \frac{\text{the mass of 1 atom of aluminium}}{\text{the mass of 1 atom of carbon}} = \frac{27}{12}$$

$$\therefore \quad \frac{\text{the mass of 2 atoms of aluminium}}{\text{the mass of 2 atoms of carbon}} = \frac{2 \times 27}{2 \times 12} = \frac{27}{12}$$

$$\text{and} \quad \frac{\text{the mass of 10 atoms of aluminium}}{\text{the mass of 10 atoms of carbon}} = \frac{10 \times 27}{10 \times 12} = \frac{27}{12}$$

$$\therefore \quad \frac{\text{the mass of N atoms of aluminium}}{\text{the mass of N atoms of carbon}} = \frac{N \times 27}{N \times 12} = \frac{27}{12}$$

Whenever the number of aluminium and carbon atoms is the same the ratio of the mass of aluminium to the mass of carbon will be 27:12.

Therefore, if we take 27g of aluminium and 12g of carbon, each sample must contain the same number of atoms.

Similarly, if we take 1g of hydrogen or 14g of nitrogen, we must have the same number of atoms in each sample as there are atoms of carbon in a 12g sample of carbon.

That is, whenever we take a sample of an element which has a mass equal to its Atomic Mass expressed in grams we have the same number of atoms in that sample as there are atoms of carbon in a 12g sample of carbon.

The very large number of atoms contained in such a sample is called Avogadro's Number.

Avogadro's Number = 6.02×10^{23} (approximately)

Similarly it can be shown that Avogadro's number of molecules of a substance has a mass which is equal to its Formula Mass expressed in grams.

Avogadro's Number (6.02×10^{23}) of particles of a substance is known as a MOLE of that substance.

DEFINITION

There are two acceptable definitions of the MOLE —

1. *A MOLE of a substance is that quantity of that substance which contains AVOGADRO'S NUMBER (6.02×10^{23}) of constituent particles.*
2. *A MOLE of a substance is that quantity of that substance which has a mass equal to its atomic (or formula) mass expressed in grams.*

FROM DEFINITION 1 OF THE MOLE

1 mole of a substance contains 6.02×10^{23} constituent particles.
$\therefore$ n moles of a subtance contains $n \times 6.02 \times 10^{23}$ constituent particles.

i.e. number of particles = number of moles $\times$ (6.02×10^{23})

Rearranging this equation gives,

$$\text{number of moles, n} = \frac{\text{number of particles}}{6.02 \times 10^{23}}$$

TYPE EXAMPLE 11: Calculate the number of moles of ammonia molecules in 60.2×10^{23} molecules of ammonia.

METHOD A (PROPORTION)

1 mole NH_3 contains 6.02×10^{23} molecules
x moles NH_3 contains 60.2×10^{23} molecules

$$\frac{x}{1} = \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}}$$

$\therefore x = 10$ moles

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3.

METHOD B (FORMULA)

number of moles of NH_3 = $n(NH_3)$
= ?
number molecules NH_3 = 60.2×10^{23}

$$n(NH_3) = \frac{\text{number molecules } NH_3}{6.02 \times 10^{23}}$$

$$= \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}}$$

= 10 moles

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3.

SET 13: NUMBER MOLECULES — MOLES

Calculate the number of moles of

1. water molecules in 602×10^{23} molecules water
2. silver atoms in 0.602×10^{23} atoms silver
3. carbon dioxide in 3.01×10^{23} molecules carbon dioxide
4. hydrogen molecules in 120.4×10^{23} molecules hydrogen
5. sulfuric acid in 1.806×10^{23} molecules sulfuric acid

TYPE EXAMPLE 12: Determine the number of molecules of oxygen in 0.4 mole oxygen (O_2).

METHOD A (PROPORTION)

1 mole O_2 contains 6.02×10^{23} molecules
0.4 mole O_2 contains x molecules

$$\frac{0.4}{1} = \frac{x}{6.02 \times 10^{23}}$$

$$\therefore x = 0.4 \times 6.02 \times 10^{23}$$
$$= 2.408 \times 10^{23} \text{ molecules}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2.

METHOD B (FORMULA)

$n(O_2) = 0.4$ mole

number molecules O_2 = ?

$$n(O_2) = \frac{\text{number molecules } O_2}{6.02 \times 10^{23}}$$

$$0.4 = \frac{\text{number molecules } O_2}{6.02 \times 10^{23}}$$

$$\therefore \text{number molecules } O_2 = 0.4 \times 6.02 \times 10^{23}$$
$$= 2.408 \times 10^{23}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2

SET 13: (continued)

Determine the number of

6. atoms of copper in 0.2 mole copper.
7. molecules of nitrogen dioxide in 5 moles nitrogen dioxide.
8. atoms of carbon in 2 moles carbon.
9. molecules of chlorine in 0.1 mole chlorine (Cl_2).
10. molecules of benzene in 2.5 moles benzene.

FROM DEFINITION 2 OF THE MOLE, to determine the mass of 1 mole of *ANY* substance simply express its formula mass in grams.

TYPE EXAMPLE 13: Determine the mass of 1 mole of —
(a) hydrogen
(b) chlorine atoms
(c) calcium oxide

(a) Hydrogen, H_2:

$$\begin{aligned} FM[H_2] &= 2 \times AM(H) \\ &= 2 \times 1 \\ &= 2 \end{aligned}$$

$\therefore$ Mass of 1 mole of hydrogen = 2g.

(b) Chlorine atoms, Cl:

$$AM[Cl] = 35.5$$

$\therefore$ Mass of 1 mole chlorine atoms = 35.5g.

(c) Calcium oxide, CaO:

$$\begin{aligned} FM[CaO] &= AM(Ca) + AM(O) \\ &= 40 + 16 \\ &= 56 \end{aligned}$$

$\therefore$ Mass of 1 mole calcium oxide = 56g.

SET 14: MOLE — MASS I

What is the mass of 1 mole of each of the following?

1. Nitrogen (gas)
2. Iron
3. Oxygen atoms
4. Silver bromide
5. Sodium hydroxide
6. Lead (II) oxide
7. Magnesium hydrogensulfate
8. Aluminium hydroxide
9. Zinc carbonate
10. Barium hydrogencarbonate

TYPE EXAMPLE 14: Determine the mass of 0.4 mole magnesium carbonate.

METHOD A (PROPORTION)

Formula: $MgCO_3$

$$\begin{aligned} FM[MgCO_3] &= 24 + 12 + 3(16) \\ &= 84 \end{aligned}$$

1 mole magnesium carbonate has mass of 84g
0.4 mole magnesium carbonate has mass of *x* g

$$\frac{x}{84} = \frac{0.4}{1}$$
$$\therefore x = 84 \times 0.4$$
$$= 33.6\text{g}$$

i.e. 0.4 mole $MgCO_3$ has a mass of 33.6g.

METHOD B (FORMULA)

A formula can be derived by simple proportion.

The mass of 1 mole of ANY substance is known as the MOLAR MASS (M) of that substance and is the formula mass expressed in grams.

If the mass of 1 mole = M,
then the mass (m) of n moles = n × mass of 1 mole
= n × M
i.e. m = n × M

This formula is more commonly referred to in the rearranged form –

$$n = \frac{m}{M}$$

where n = number of moles
m = mass of substance (g)
M = molar mass (g)

Note: Strictly speaking the units of M should be $g.mole^{-1}$

Appling this formula to *TYPE EXAMPLE 14.*

METHOD B (FORMULA)

Formula: $MgCO_3$

$$FM[MgCO_3] = 24 + 12 + 48$$
$$= 84$$

$n\ (MgCO_3) = 0.4$ mole
$m\ (MgCO_3) = ?$
$M\ (MgCO_3) = 84\text{g}$

$$n = \frac{m}{M}$$
$$0.4 = \frac{m}{84}$$
$$\therefore m = 0.4 \times 84$$
$$= 33.6\text{g}$$

Again, mass of 0.4 mole $MgCO_3$ is 33.6g.

SET 15: MOLE — MASS II

Determine the mass of —

1. 2 moles aluminium
2. 5 moles oxygen gas
3. 0.2 mole zinc oxide
4. 1.5 moles barium carbonate
5. 0.6 mole copper (II) chloride
6. 0.1 mole iron (II) nitrate
7. 3.2 moles potassium sulfate
8. 2.5 moles sodium hydrogencarbonate
9. 0.5 mole mercury (II) nitrate
10. 0.4 mole iron (II) sulfide

TYPE EXAMPLE 15: Calculate how many moles of water are in 72g water.

Formula: H_2O

$FM[H_2O] = 2(1) + 16 = 18$

Mass of 1 mole water, $M(H_2O) = 18g$

METHOD A (PROPORTION)

1 mole H_2O has a mass of 18g
x mole H_2O has a mass of 72g

$$\frac{x}{1} = \frac{72}{18}$$

$$\therefore x = 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O

METHOD B (FORMULA)

$M(H_2O) = 18g$
$m(H_2O) = 72g$
$n(H_2O) = ?$

$$n = \frac{m}{M}$$
$$= \frac{72}{18}$$
$$= 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O.

SET 16: MASS — MOLE

Determine the number of moles of

1. magnesium in 48g magnesium
2. hydrogen in 1g hydrogen gas
3. carbon dioxide in 4.4g carbon dioxide
4. calcium sulfate in 27.2g calcium sulfate
5. potassium hydroxide in 22.4g potassium hydroxide
6. zinc nitrate in 947g zinc nitrate
7. ammonium carbonate in 288g ammonium carbonate
8. lead (II) chloride in 55.6g lead (II) chloride
9. sodium sulfate in 85.2g sodium sulfate
10. silver chloride in 114.8g silver chloride

TYPE EXAMPLE 16; If 1.5 moles of benzene has a mass of 117g, determine the formula mass of benzene.

METHOD A (PROPORTION)

1.5 moles has mass of 117g.
1 mole benzene has mass of xg

$$\frac{x}{117} = \frac{1}{1.5}$$
$$\therefore x = \frac{117}{1.5}$$
$$= 78g$$

From the definition of a mole, the mass of 1 mole of any substance is its formula mass expressed in grams.

$\therefore$ Formula mass of benzene is 78

METHOD B (FORMULA)

n (benzene) = 1.5 moles
m (benzene) = 117g
M (benzene) = ?

$$n = \frac{m}{M}$$
$$1.5 = \frac{117}{M}$$
$$\therefore M = \frac{117}{1.5}$$
$$= 78g$$

i.e. Formula mass of benzene is 78.

SET 17: FORMULA MASS

Determine the formula mass of the substance in each of the questions below, if

1. 3 moles acetylene has a mass of 78g
2. 2 moles phenol has a mass of 188g
3. 0.1 mole calcium hypochlorite has a mass of 14.3g
4. 0.2 mole methanol has a mass of 6.4g
5. 1.5 moles ethylene has a mass of 42g
6. the mass of 5 moles propane is 220g
7. the mass of 0.4 mole ethanol is 18.4g
8. the mass of 2.5 moles of a gas is 115g
9. the mass of 0.1 mole ethane is 3g
10. the mass of 0.3 mole of a silvery metal is 15.6g.

TYPE EXAMPLE 17: Determine the number of moles of atoms of hydrogen, sulfur and oxygen in 1 mole sulfuric acid.

Formula: H_2SO_4

Examination of the formula shows 1 molecule of H_2SO_4 contains–
(a) 2 atoms H
(b) 1 atom S
(c) 4 atoms O

∴Avogadro's Number of molecules of H_2SO_4 must contain —

(a) 2 × Avogadro's Number of H atoms.
(b) 1 × Avogadro's Number of S atoms.
(c) 4 × Avogadro's Number of O atoms.

i.e. 1 mole H_2SO_4 contains
(a) 2 moles H atoms
(b) 1 mole S atoms
(c) 4 moles O atoms

TYPE EXAMPLE 18: Find the number of moles of potassium ions in 0.2 mole potassium sulfate.

Formula: K_2SO_4

CONCENTRATION OF A SOLUTION

A *solution* is formed by dissolving one substance *(solute)* in another *(solvent)*, usually a liquid. Water is the most common solvent.

The amount of solute that is dissolved per unit volume of solution is called the *concentration* of the solution.

Concentration (C) is commonly expressed as
— moles of solute (n) per litre of solution (V)

$$C = \frac{n}{V} \quad \text{units: mol } L^{-1}$$

— mass in grams of solute (m) per litre of solution(V)

$$C = \frac{m}{V} \quad \text{units: } gL^{-1}$$

A *saturated solution* contains as much dissolved solute as is possible in that volume of solvent at that temperature. The concentration of a saturated solution is a measure of the *solubility* of the solute in that solvent at that temperature. Different solutes have different solubilities in the same solvent at the same temperature.

TYPE EXAMPLE 22: The concentration of a silver nitrate solution is 0.1 mol L^{-1}. What is its concentration in grams per litre?

METHOD A (PROPORTION)

1L of solution contains 0.1 mol $AgNO_3$

$FM\ (AgNO_3) = 108 + 14 + 3(16)$
$= 170$

1 mole $AgNO_3$ has a mass of 170g
0.1 mole $AgNO_3$ has a mass of xg

$$\frac{x}{170} = \frac{0.1}{1}$$

$x = 0.1 \times 170$
$= 17g$

i.e. the concentration of $AgNO_3$ solution is $17gL^{-1}$.

METHOD B (FORMULA)

1L of solution contains 0.1 mol $AgNO_3$

$n(AgNO_3) = 0.1$
$m(AgNO_3) = ?$
$M(AgNO_3) = 170g$

$$n = \frac{m}{M}$$

$$0.1 = \frac{m}{170}$$

$m = 0.1 \times 170$
$= 17g$

$FM(AgNO_3) = 108 + 14 + 48$
$= 170$

i.e. the concentration of $AgNO_3$ solution is $17gL^{-1}$.

TYPE EXAMPLE 23: The concentration of a potassium hydroxide solution is 84gL^{-1}. What is its concentration in mol L^{-1}?

METHOD A (PROPORTION)

1L of solution contains 84g KOH

FM(KOH) = 39 + 16 + 1
= 56

1 mole KOH has a mass of 56g
x mole KOH has a mass of 84g

$$\frac{x}{1} = \frac{84}{56}$$

$$x = 1.5 \text{ mol}$$

i.e. the concentration of KOH solution is 1.5 mol L^{-1}.

METHOD B (FORMULA)

1 L of solution contains 84g KOH.

n(KOH) = ?
m(KOH) = 84g
M(KOH) = 56g

$$n = \frac{m}{M}$$

$$= \frac{84}{56}$$

$$= 1.5 \text{ mol}$$

FM(KOH) = 39 + 16 + 1
= 56

i.e. the concentration of KOH solution is 1.5 mol L^{-1}

SET 22: CONCENTRATION

1. The concentration of an aqueous calcium sulfate solution is 27.2gL^{-1}. What is its concentration in mol L^{-1}?
2. The concentration of an aqueous lead (II) nitrate solution is 0.50 mol L^{-1}. What is its concentration in g L^{-1}?
3. If 50g of sodium hydroxide is dissolved in water and the volume of solution made up to 1 L, what is the concentration of sodium hydroxide solution both in g L^{-1} and mol L^{-1}?
4. If 0.25 mol of sodium nitrate is dissolved in water and the volume of solution made up to 1 L, what is the concentration of sodium nitrate solution both in mol L^{-1} and g L^{-1}?
5. The solubilities of iodine and benzoic acid (C_6H_5COOH) in water at 15^{0}C are 0.0011 mol L^{-1} and 0.021 mol L^{-1}, respectively. What are the concentrations of saturated solutions of iodine and benzoic acid, in g L^{-1}, at 15^{0}C?
6. The solubilities of sodium chloride and calcium hydroxide in water at 20^{0}C are 360 g L^{-1} and 1.6 g L^{-1}, respectively. What are the concentrations of saturated solutions of sodium chloride and calcium hydroxide, in mol L^{-1}, at 20^{0}C?

7. Like most solids, the solubility of patassium chloride in water increases as the temperature increases. At 20°C its solubility is 340g L^{-1}, whereas at 100°C its solubility is 567 g L^{-1}. What is the concentration of a saturated solution of potassium chloride, in mol L^{-1}, at both 20°C and 100°C?
8. Like most gases the solubility of carbon dioxide in water decreases as the temperature increases. At 20°C its solubility is 1.5 g L^{-1} whereas at 40°C it is 0.9 g L^{-1}. Calculate the solubility of carbon dioxide in mol L^{-1}, at both temperatures.
9. Like all gases the solubility of oxygen in water increases as the pressure of the gas above the liquid increases. At 106.4 kPa and 26°C its solubility is 0.0012 mol L^{-1} whereas it is 0.0023 mol L^{-1} at 199.6 kPa and 26°C. Determine the solubility of oxygen, in g L^{-1}, at both pressures.
10. Plot the solubility curve for copper (II) sulfate pentahydrate. Use graph paper and sketch in the line-of-best-fit through the plotted points.

Temperature (°C)	0	10	20	30	40	50	60	70	80	90	100
Solubility (g L^{-1})	150	180	210	240	280	340	400	480	580	720	1000

From the graph find:

(a) the temperature at which the solubility is 360 g L^{-1}.
(b) the solubility of this salt at (i) 75°C (ii) 45°C.
(c) the mass of solute deposited if one litre of saturated solution at 75°C is cooled to 45°C.
(d) the solubility of this salt, in mol L^{-1}, at 25°C.
(e) if one litre of hot solution contained 250g of this salt, at what temperature would crystallization begin (indicating the solution is saturated)?

CHAPTER 13: Problems Involving Equations.

In any chemical reaction if the quantity of one substance (either reacted or produced) is known and a balanced chemical equation is written, the quantity of any other reactant (or product) can be determined.

The quantity of the substance known may be either in

(a) moles, or
(b) mass, or
(c) volume

The BALANCED chemical equation is the key to the solution.

Consider the simple reaction between hydrogen and nitrogen to produce ammonia

Equation: $N_2 + 3H_2 \longrightarrow 2NH_3$

The information given by this equation is that:

1 molecule N_2 reacts with 3 molecules H_2 to produce 2 molecules NH_3

Similarly —

$(6.02 \times 10^{23}$ molecules) N_2 reacts with $3(6.02 \times 10^{23}$ molecules) H_2 to produce $2(6.02 \times 10^{23}$ molecules) NH_3.

From the definition of the mole, 6.02×10^{23} molecules of any substance are contained in 1 mole of that substance.

Hence: 1 mole N_2 reacts with 3 moles H_2 to produce 2 moles NH_3.

Summarizing:

$$N_2 \quad + \quad 3H_2 \longrightarrow 2NH_3$$

1 molecule	3 molecules	2 molecules
(6.02×10^{23}) molecules	$3(6.02 \times 10^{23}$ molecules)	$2(6.02 \times 10^{23}$ molecules)
1 mole	3 moles	2 moles

MOLE-MOLE DETERMINATIONS

TYPE EXAMPLE 28: Determine the number of moles of chlorine required to react with 0.2 mole sodium to produce sodium chloride.

Equation: $2Na + Cl_2 \longrightarrow 2NaCl$

METHOD A (PROPORTION)

$$2Na + Cl_2 \longrightarrow 2NaCl$$
$$2 \text{ mole} \quad 1 \text{ mole}$$

2 moles Na reacts with 1 mole Cl_2 (from equation)
0.2 mole Na reacts with x mole Cl_2

$$\frac{x}{1} = \frac{0.2}{2}$$
$$\therefore x = 0.1 \text{ mole}$$

i.e. 0.2 mole sodium reacts with 0.1 mole chlorine.

An alternative method (from formula) may be preferred to the proportion method. Its application to the solution of any problem based upon an equation is according to the following steps:

STEP 1: Write down the balanced equation and identify the known and unknown substances.

STEP 2: Convert (if necessary) the known quantity of the substance into MOLES.

STEP 3: From the equation write down the mole ratio of unknown to the known

$$\frac{n \text{ (unknown)}}{n \text{ (known)}} = \frac{\text{coefficient of unknown in the balanced equation}}{\text{coefficient of the known in the balanced equation}}$$

STEP 4: Calculate the number of moles of the unknown substance.

STEP 5: Convert the mole quantity determined in Step 4 to the required units (if necessary).

METHOD B (FORMULA)

STEP 1:

$$\underset{\text{(known)}}{2Na} + \underset{\text{(unknown)}}{Cl_2} \longrightarrow 2NaCl$$

(STEP 2: not required)

STEP 3:

$$\frac{n(Cl_2)}{n(Na)} = \frac{1}{2}$$

STEP 4:

$n(Cl_2) = ?$

$n(Na) = 0.2$ mole

$$\frac{n(Cl_2)}{0.2} = \frac{1}{2}$$

$$\therefore n(Cl_2) = 0.2 \times \tfrac{1}{2}$$

$$= 0.1 \text{ mole}$$

(STEP 5: not required)

i.e. 0.2 mole Na requires 0.1 mole Cl_2.

SET 28: MOLE-MOLE DETERMINATIONS

1. Determine the number of moles of magnesium oxide produced when 2 moles oxygen is reacted as in the equation:

$$2Mg + O_2 \longrightarrow 2MgO$$

2. Mercuric oxide when heated produces mercury and oxygen according to:

$$2HgO \longrightarrow 2Hg + O_2$$

If 0.4 mole mercuric oxide is decomposed how many moles of —
(a) mercury, (b) oxygen are produced?

3. Find the number of moles of hydrochloric acid required to react with 0.1 mole zinc metal:

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$$

4. Find (a) the number of moles of calcium oxide
(b) the number of moles of carbon dioxide
produced when 0.30 mole calcium carbonate is decomposed by strong heating according to:

$$CaCO_3 \longrightarrow CaO + CO_2$$

5. 1.5 moles potassium nitrate are heated:

$$2KNO_3 \longrightarrow 2KNO_2 + O_2$$

How many moles of —
(a) potassium nitrite are left over, and
(b) oxygen are evolved?

6. The electrolysis of water can be represented by the equation:

$$2H_2O \longrightarrow 2H_2 + O_2$$

If 0.5 mole oxygen has been produced determine the moles of water electrolysed.

7. How many moles of sodium carbonate, when reacted with excess sulfuric acid, will produce 0.22 mole carbon dioxide?

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

8. A piece of copper, when placed in a silver nitrate solution, deposits silver from the solution:

$$Cu + 2AgNO_3 \longrightarrow Cu(NO_3)_2 + 2Ag$$

Determine the moles of copper consumed if 0.04 mole silver is deposited.

9. Sodium when added to water reacts rapidly:

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

If 0.1 mole hydrogen gas is evolved, determine the moles of sodium metal used.

10. Aluminium burns according to:

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

If 0.2 mole aluminium oxide is produced, how many moles of aluminium was burnt?

MOLE-MASS DETERMINATIONS:

TYPE EXAMPLE 29: Determine the mass of hydrogen evolved when 0.1 mole calcium is reacted with water.

METHOD A (PROPORTION)

$$\underset{1 \text{ mole}}{Ca} + 2H_2O \longrightarrow Ca(OH)_2 + \underset{1 \text{ mole}}{H_2}$$

1 mole calcium produces 1 mole hydrogen (from equation) $FM[H_2] = 2$

$\therefore$ 1 mole calcium produces 2g hydrogen

0.1 mole calcium produces xg hydrogen

$$\frac{x}{2} = \frac{0.1}{1}$$

$$\therefore x = \frac{2 \times 0.1}{1}$$

$$= 0.2 \text{ g}$$

i.e. 0.1 mole calcium produces 0.2g hydrogen.

METHOD B (FORMULA)

Step 1:

$$\underset{\text{(known)}}{Ca} + 2H_2O \longrightarrow Ca(OH)_2 + \underset{\text{(unknown)}}{H_2}$$

$FM(H_2) = 2$

(Step 2: not required)

Step 3:

$$\frac{n(H_2)}{n(Ca)} = \frac{1}{1}$$

Step 4:

$n(Ca) = 0.1$ mole

$$\frac{n(H_2)}{0.1} = \frac{1}{1}$$

$$\therefore n(H_2) = \frac{0.1}{1} \times \frac{1}{1}$$

$$= 0.1 \text{ mole}$$

Step 5:

$n(H_2) = 0.1$ mole
$m(H_2) = ?$
$M(H_2) = 2g$

$$n = \frac{m}{M}$$

$$0.1 = \frac{m}{2}$$

$$\therefore m = 2 \times 0.1$$

$$= 0.2g$$

i.e. 0.1 mole calcium produces 0.2g hydrogen.

SET 29: MOLE-MASS DETERMINATIONS

1. Determine the mass of sulfur dioxide produced when 0.2 mole sulfur is burnt in excess oxygen.

$$S_8 + 8O_2 \longrightarrow 8SO_2$$

2. (a) How many grams of sodium carbonate remain when 0.1 mole sodium hydrogencarbonate is heated?

$$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$

(b) How many grams of carbon dioxide is produced in (a)?

3. 0.25 mole sodium hydroxide is neutralized by dilute sulfuric acid. Find the mass of sodium sulfate produced.

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

4. Potassium chlorate, when heated, liberates oxygen and leaves potassium chloride.

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

If 0.2 mole potassium chlorate is heated determine the —
(a) mass of potassium chloride left.
(b) mass of oxygen liberated.

5. Methane burns in air (oxygen) according to —

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

If 0.02 mole methane is used determine the mass of water vapour produced.

MASS-MOLE DETERMINATIONS:

TYPE EXAMPLE 30: Find the mass of sodium chloride which must be added to lead(II) nitrate solution to deposit 0.05 mole lead(II) chloride.

METHOD A (PROPORTION)

$$Pb(NO_3)_2 + 2NaCl \longrightarrow PbCl_2 + 2NaNO_3$$
$$\text{2 moles} \quad \text{1 mole}$$

$$FM[NaCl] = 23 + 35.5 = 58.5$$

2 mole NaCl produces 1 mole $PbCl_2$ (from equation)
$\therefore$ (2 × 58.5)g NaCl produces 1 mole $PbCl_2$
xg NaCl produces 0.05 mole $PbCl_2$

$$\frac{x}{2 \times 58.5} = \frac{0.05}{1}$$
$$\therefore x = 0.05 \times (2 \times 58.5)$$
$$= 5.85g$$

i.e. 0.05 mole $PbCl_2$ deposited by 5.85g NaCl.

METHOD B (FORMULA)

Step 1:

$$Pb(NO_3)_2 + 2NaCl \longrightarrow PbCl_2 + 2NaNO_3$$
$$\text{(unknown)} \quad \text{(known)}$$

$$FM(NaCl) = 23 + 35.5 = 58.5$$

(Step 2: Not required.)

Step 3:

$$\frac{n(NaCl)}{n(PbCl_2)} = \frac{2}{1}$$

Step 4:

$n(NaCl) = ?$
$n(PbCl_2) = 0.05$ mole

$$\frac{n(NaCl)}{0.05} = \frac{2}{1}$$
$$\therefore n(NaCl) = \frac{2}{1} \times 0.05$$
$$= 0.1 \text{ mole}$$

Step 5:

n(NaCl) = 0.1 mole
m(NaCl) = ?
M(NaCl) = 58.5g

$$n = \frac{m}{M}$$

$$0.1 = \frac{m}{58.5}$$

$$\therefore m = 0.1 \times 58.5g$$

$$= 5.85g$$

i.e. 0.05 mole $PbCl_2$ deposited by 5.85g NaCl.

SET 30: MOLE-MASS DETERMINATIONS

1. What mass of magnesium must be added to an excess of hydrochloric acid to liberate 0.2 mole hydrogen?

$$Mg + 2HCl \longrightarrow MgCl_2 + H_2$$

2. Determine the mass of (a) ammonia, and (b) sulfuric acid required to produce 2 moles ammonium sulfate.

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

3. How many grams of manganese dioxide is required to produce 0.1 mole chlorine on reaction with concentrated hydrochloric acid?

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

4. 0.05 mole copper(II) oxide remains when a sample of copper(II) carbonate is heated.

$$CuCO_3 \longrightarrow CuO + CO_2$$

Determine the mass of
(a) copper(II) carbonate required, and
(b) carbon dioxide produced.

5. 0.2 mole oxygen is liberated when water is added to sodium peroxide.

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

Determine
(a) the mass of sodium peroxide used, and
(b) the number of moles of sodium hydroxide remaining in the solution.

MASS-MOLE DETERMINATIONS:

TYPE EXAMPLE 31: Find the number of moles of ammonia produced when 18.5g calcium hydroxide is heated with sufficient ammonium chloride.

METHOD A (PROPORTION)

$$Ca(OH)_2 + 2NH_4Cl \longrightarrow CaCl_2 + 2H_2O + 2NH_3$$

1 mole — 2 moles

$$FM[Ca(OH)_2] = 40 + 2(16) + 2(1)$$
$$= 74$$

1 mole $Ca(OH)_2$ produces 2 moles NH_3 (from equation)
$\therefore$ 74g $Ca(OH)_2$ produces 2 moles NH_3
18.5g $Ca(OH)_2$ produces x moles NH_3

$$\frac{x}{2} = \frac{18.5}{74}$$
$$\therefore x = \frac{2 \times 18.5}{74}$$
$$= 0.5 \text{ mole}$$

i.e. 18.5g $Ca(OH)_2$ produces 0.5 mole NH_3

METHOD B (FORMULA)

Step 1:

$$Ca(OH)_2 + 2NH_4Cl \longrightarrow CaCl_2 + 2H_2O + 2NH_3$$

(known) — (unknown)

$$FM[Ca(OH)_2] = 40 + 2(16) + 2(1)$$
$$= 74$$

Step 2:

$n[Ca(OH)_2] = ?$
$m[Ca(OH)_2] = 18.5g$
$M[Ca(OH)_2] = 74g$

$$n = \frac{m}{M}$$
$$= \frac{18.5}{74}$$
$$= 0.25 \text{ mole}$$

Step 3:

$$\frac{n(NH_3)}{n[Ca(OH)_2]} = \frac{2}{1}$$

Step 4:

$n(NH_3) = ?$
$n[Ca(OH)_2] = 0.25$ mole

$$\frac{n(NH_3)}{0.25} = \frac{2}{1}$$
$$\therefore n(NH_3) = 2 \times 0.25$$
$$= 0.5 \text{ mole}$$

i.e. 18.5g $Ca(OH)_2$ produces 0.5 mole NH_3

SET 31: MASS-MOLE DETERMINATIONS

1. Determine the number of moles of zinc oxide produced when 13.08g zinc is heated in sufficient oxygen.

$$2Zn + O_2 \longrightarrow 2ZnO$$

2. How many moles of calcium carbonate will be deposited when 53g sodium carbonate is added to a solution of calcium chloride?

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3(s) + 2NaCl$$

3. Find the number of moles of (a) ammonia, and (b) carbon dioxide produced when 4.8g ammonium carbonate is decomposed.

$$(NH_4)_2CO_3 \longrightarrow 2NH_3 + H_2O + CO_2$$

4. Calculate the number of moles of sulfuric acid required to neutralize 32g sodium hydroxide.

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

5. Determine the number of moles of sulfur trioxide formed when 16g sulfur dioxide is oxidised.

$$2SO_2 + O_2 \longrightarrow 2SO_3$$

6. (a) How many moles of sodium nitrate must be heated to produce 4g oxygen?

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

(b) How many moles of sodium nitrite will remain?

7. 3g carbon is produced when magnesium is burnt in carbon dioxide.

$$2Mg + CO_2 \longrightarrow 2MgO + C$$

How many moles of magnesium is required?

8. Determine the number of moles of water which must be electrolysed to produce 128g oxygen.

$$2H_2O \longrightarrow 2H_2 + O_2$$

9. 54.56g zinc chloride is recovered when zinc oxide is neutralized with sufficient hydrochloric acid.

$$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$$

How many moles of zinc oxide is neutralized?

10. If 34.44g silver chloride is deposited when sufficient silver nitrate is added to a solution of aluminium chloride, determine the number of moles of aluminium chloride required.

$$AlCl_3 + 3AgNO_3 \longrightarrow Al(NO_3)_3 + 3AgCl(s)$$

MASS-MASS DETERMINATIONS:

TYPE EXAMPLE 32: What mass of hydrogen is produced when 6g magnesium is added to excess sulfuric acid?

METHOD A (PROPORTION)

$$\underset{1\text{ mole}}{Mg} + H_2SO_4 \longrightarrow MgSO_4 + \underset{1\text{ mole}}{H_2}$$

FM (Mg) = 24
FM (H_2) = 2

1 mole magnesium produces 1 mole hydrogen (from equation)
∴ 24g magnesium produces 2g hydrogen
6g magnesium produces xg hydrogen

$$\frac{x}{2} = \frac{6}{24}$$

$$\therefore x = \frac{2 \times 6}{24}$$

$$= 0.5g$$

i.e. 6g Mg produces 0.5g H_2.

METHOD B (FORMULA)

Step 1:

$$\underset{\text{(known)}}{Mg} + H_2SO_4 \longrightarrow MgSO_4 + \underset{\text{(unknown)}}{H_2}$$

FM(Mg) = 24
FM(H_2) = 2

Step 2:

m(Mg) = 6g
n(Mg) = ?
M(Mg) = 24g

$$n = \frac{m}{M}$$

$$= \frac{6}{24}$$

$$= 0.25 \text{ mole}$$

Step 3:

$$\frac{n(H_2)}{n(Mg)} = \frac{1}{1}$$

Step 4:

n(Mg) = 0.25 mole
$n(H_2)$ = ?

$$\frac{n(H_2)}{0.25} = \frac{1}{1}$$

$$\therefore n(H_2) = 0.25 \text{ mole}$$

Step 5:

$n(H_2) = 0.25$ mole

$m(H_2) = ?$

$M(H_2) = 2g$

$$n = \frac{m}{M}$$

$$0.25 = \frac{m}{2}$$

$$\therefore m = 0.25 \times 2$$

$$= 0.5g$$

i.e. 6g Mg produces 0.5g H_2.

SET 32: MASS-MASS DETERMINATIONS I

1. 25g calcium carbonate is strongly heated.

$$CaCO_3 \longrightarrow CaO + CO_2$$

Determine the mass of calcium oxide produced.

2. In an experiment 3.6g water is electrolysed.

$$2H_2O \longrightarrow 2H_2 + O_2$$

Find the mass of
(a) hydrogen, and
(b) oxygen produced.

3. 20g sodium hydroxide is neutralized by dilute hydrochloric acid.

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

How many grams of sodium chloride can be recovered?

4. Determine the mass of calcium sulfate produced when 5.6g calcium oxide is neutralized by sulfuric acid.

$$CaO + H_2SO_4 \longrightarrow CaSO_4 + H_2O$$

5. 6.9g sodium metal is reacted with water. Find the mass of sodium hydroxide produced.

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

6. 420g sodium hydrogencarbonate is strongly heated.

$$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$

Determine the mass of —
(a) sodium carbonate left, and
(b) carbon dioxide produced.

7. Determine the mass of hydrochloric acid required to completely react with 16.8g magnesium carbonate.

$$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

8. 6.35g copper is dissolved when placed in a solution of silver nitrate.

$$Cu(s) + 2AgNO_3 \longrightarrow Cu(NO_3)_2 + 2Ag(s)$$

How many grams of
(a) silver is deposited, and
(b) copper(II) nitrate can be recovered?

9. Sufficient potassium carbonate solution is added to 34.1g zinc chloride to completely precipitate the zinc as zinc carbonate.

$$ZnCl_2 + K_2CO_3 \longrightarrow ZnCO_3(s) + 2KCl$$

What mass of zinc carbonate is deposited?

10. Determine the mass of
(a) sodium nitrite left, and
(b) oxygen gas evolved
when 17g sodium nitrate is strongly heated

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2(g)$$

SET 33: MASS-MASS DETERMINATIONS II

1. Determine the mass of potassium chlorate which must be decomposed by heating to produce 9.6g oxygen.

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

2. 10.8g aluminium powder is burnt in air.

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Find the mass of aluminium oxide produced.

3. 66g ammonium sulfate is produced when ammonia gas is bubbled through a sulfuric acid solution.

$$2NH_3(g) + H_2SO_4 \longrightarrow (NH_4)_2SO_4.$$

What mass of ammonia is required?

4. Calculate the mass of hydrogen peroxide which upon catalytic decomposition will produce 4g oxygen.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

5. Determine the mass of sodium carbonate which must be added to a calcium nitrate solution to precipitate all of the calcium as 8g of calcium carbonate.

$$Na_2CO_3 + Ca(NO_3)_2 \longrightarrow CaCO_3(s) + 2NaNO_3$$

6. Find the mass of chlorine produced when 4.35g manganese dioxide is reacted with concentrated hydrochloric acid.

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

7. What mass of barium chloride is produced in the reaction between hydrochloric acid and barium carbonate which produces 8.8g carbon dioxide?

$$BaCO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + CO_2(g)$$

ANSWERS — PART 2

SET 1

1. 18	**2.** 44	**3.** 28	**4.** 71	**5.** 80
6. 46	**7.** 98	**8.** 48	**9.** 58	**10.** 46

SET 2

1. HCl, 36.5	**2.** CaO, 56	**3.** ZnS, 97.4
4. $AlCl_3$, 133.5	**5.** MgO, 40	**6.** Na_2S, 78
7. KBr, 119	**8.** CuO, 79.5	**9.** $BaCl_2$, 208
10. Ag_2S, 248		

SET 3

1. Na_2CO_3, 106	**2.** $Pb(NO_3)_2$, 331	**3.** $Mg(OH)_2$, 58
4. $CuCO_3$, 123.5	**5.** $Fe(OH)_2$, 90	**6.** NH_4Cl, 53.5
7. $Ba(NO_3)_2$, 261	**8.** $KHCO_3$, 100	**9.** $AgNO_3$, 170
10. $Ca_3(PO_4)_2$, 310		

SET 4

1. $Pb(OH)_2$, 241	**2.** K_2SO_4, 174	**3.** $(NH_4)_2S$, 68
4. $Al_2(CO_3)_3$, 234	**5.** Mg_3N_2, 100	**6.** $(NH_4)_2SO_4$, 132
7. $Zn(HCO_3)_2$, 187.4	**8.** FeO, 72	**9.** $Ca(HSO_4)_2$, 234
10. Na_3PO_4, 164		

SET 5

1. 203	**2.** 154.4	**3.** 278	**4.** 286	**5.** 482

SET 6

1. C 27.3%; O 72.7%	**2.** N 82.4%; H 17.6%
3. Ca 40%; C 12%; O 48%	**4.** Mg 25.3%; Cl 74.7%
5. Ag 63.5%; N 8.2%; O 28.2%	

SET 7

1. 11.1%	**2.** 40%	**3.** 75%	**4.** 57.1%	**5.** 32.4%
6. 18.9%	**7.** 74.5%	**8.** 18.7%	**9.** 92.6%	**10.** 23.5%

SET 8

1. 72.9%	**2.** 22.5%	**3.** 47.7%	**4.** 84.2%	**5.** 83.6%

SET 9

1. 14.8%	**2.** 51.2%	**3.** 21.1%	**4.** 39.9%	**5.** 22.4%

SET 10

1. 4g	**2.** 6.4g	**3.** 2.8g	**4.** 254g	**5.** 130.8g
6. 68.5g	7. 1.2g	**8.** 6.9g	**9.** 1g	**10.** 8.4g

SET 11

1. 14g	**2.** 117g	**3.** 11.6g	**4.** 51g	**5.** 26.4g
6. 46.1g	7. 116.5g	**8.** 44.7g	**9.** 35.6g	**10.** 8.7g

SET 12

1(a) $CaCl_2$, 111 **(b)** NH_4NO_3, 80

2(a) $Ba(HSO_4)_2$, 331 **(b)** $Zn(OH)_2$, 99.4

3. Mg, 20%; S, 26.7%; O, 53.3%

4. 8.7%		
5. 62.9%	**6.** 6.4g	7. 32.7g
8. 39g	**9.** 89g	**10.** 54.25g

SET 13

1. 100	**2.** 0.1	3. 0.5	**4.** 20	**5.** 0.3
6. 1.204×10^{23}		7. 30.1×10^{23}		**8.** 12.04×10^{23}
9. 0.602×10^{23}		**10.** 15.05×10^{23}		

SET 14

1. 28g	**2.** 56g	**3.** 16g	**4.** 188g	**5.** 40g
6. 223g	7. 218g	**8.** 78g	**9.** 125.4g	**10.** 259g

SET 15

1. 54g	**2.** 160g	**3.** 16.28g	**4.** 295.5g	**5.** 80.7g
6. 18g	7. 556.8g	**8.** 210g	**9.** 162.5g	**10.** 35.2g

SET 16

1. 2	**2.** 0.5	**3.** 0.1	**4.** 0.2	**5.** 0.4
6. 5	7. 3	**8.** 0.2	**9.** 0.6	**10.** 0.8

SET 17

1. 26	**2.** 94	**3.** 143	**4.** 32	**5.** 28
6. 44	7. 46	**8.** 46	**9.** 30	**10.** 52

SET 18

1. 0.2 **2.** H, 2; N, 2; O, 6 **3.** Cu, 2.5; S, 2.5; O, 10

4. Ca^{2+}, 0.4; OH^-, 0.8 **5.** 0.602×10^{23} of each

6. H, 36.12×10^{23}; S, 18.06×10^{23}

7.(a) N, 0.2; H, 0.6 **(b)** NH_3, 1.204×10^{23}; N, 1.204×10^{23}; H, 3.612×10^{23}

8. N, 1; H, 4; C, 0.5; O, 1.5

SET 19

1. 0.1	**2.** 0.4	**3.** 2	**4.** 1.5	**5.** 0.4
6. 3	**7.** 0.15	**8.** 0.75	**9.** 0.4	**10.** 1.4

SET 20

1. 32g	**2.** 4g	**3.** 14g	**4.** 6.4g	**5.** 22.4g
6. 48g	**7.** 11.7g	**8.** 2 010g	**9.** 12.8g	**10.** 48g

SET 21

1.(a) 40g (b) 73g **2.(a)** 0.602×10^{23} **(b)** 36.12×10^{23}

3.(a) 0.2 **(b)** 0.4 **4.** 0.5 **5.(a)** 28g **(b)** 14g

6. 249.5g **7.(a)** 46.4g **(b)** 390g **8.** 0.3

9.(a) 0.1 **(b)** 1 **10.** 60g **11.(a)** 0.2 **(b)** 2.8g

12. 70 **13.(a)** 255g **(b)** 53g **14.** 0.29

15. 5.36 **16.** 0.4 **17.(a)** 106.5g **(b)** 7.1g

18. 16.8g **19.** 0.4 **20.** 64

SET 22

1. 0.2 mol L^{-1} **2.** 165.5 g L^{-1} **3.** 50 g L^{-1}, 1.25 mol L^{-1} **4.** 0.25 mol L^{-1} 21.25 g L^{-1} **5.** 0.279 g L^{-1}, 2.56 g L^{-1} **6.** 6.15 mol L^{-1}, 0.0216 mol L^{-1} **7.**4.56 mol L^{-1}, 7.61 mol L^{-1} **8.** 0.0341 mol L^{-1},0.0204 mol L^{-1} **9.** 0.0384 g L^{-1}, 0.0736 g L^{-1} **10.** (a) about 53^0 (b) (i) about 525 g L^{-1} (ii) about 310gL^{-1} (c) 215g (d) about 0.88 mol L^{-1} (e) about 32^0C

SET 23

1. 156.8L	**2.** 56L	**3.** 0.896L	**4.** 224L	**5.** 0.0224L (22.4mL)
6. 0.1	**7.** 0.05	**8.** 8	**9.** 70	**10.** 30

SET 24

1. 44.8L	**2.** 5.6L	**3.** 8.96L	**4.** 89.6L	**5.** 1.12L
6. 4.48L	**7.** 33.6L	**8.** 2.8L	**9.** 24.64L	**10.** 56L

SET 25

1. 11g	**2.** 3.55g	**3.** 0.6g	**4.** 160g	**5.** 10g
6. 40g	**7.** 127.75g	**8.** 2.55g	**9.** 34.5g	**10.** 3.2g

SET 26

1. 53.76L	**2.** 2.5	**3.** 4.48L	**4.** 1.4L	**5.** 38.4g
6.(a)7.5 **(b)** 330g	**7.** 0.4	**8.** 15.68L	**9.** 0.75	**10.** 45.6g

SET 27 qualitative, no answers

SET 28

1. 4 **2.(a)** 0.4 **(b)** 0.2 **3.** 0.2 **4.(a)** 0.3 **(b)** 0.3 **5.(a)** 1.5 **(b)** 0.75
6. 1 **7.** 0.22 **8.** 0.02 **9.** 0.2 **10.** 0.4

SET 29

1. 102.4g **2.(a)** 5.3g **(b)** 2.2g **3.** 17.75g
4.(a) 14.9g **(b)** 9.6g **5.** 0.72g

SET 30

1. 4.8g **2.(a)** 68g **(b)** 196g **3.** 8.7g
4.(a) 6.175g **(b)** 2.2g **5.(a)** 31.2g **(b)** 0.8

SET 31

1. 0.2 **2.** 0.5 **3.(a)** 0.1 **(b)** 0.05
4. 0.4 **5.** 0.25 **6.(a)** 0.25 **(b)** 0.25
7. 0.5 **8.** 8 **9.** 0.4
10. 0.08

SET 32

1. 14g **2.(a)** 0.4g **(b)** 3.2g **3.** 29.25g
4. 13.6g **5.** 12g **6.(a)** 265g **(b)** 110g
7. 14.6g **8.(a)** 21.6g **(b)** 18.75g **9.** 31.35g
10.(a) 13.8g **(b)** 3.2g

SET 33

1. 24.5g **2.** 20.4g **3.** 17g **4.** 8.5g **5.** 8.48g
6. 3.55g **7.** 41.6g **8.** 44kg **9.** 45g
10.(a) 21.7g **(b)** 20.1g

SET 34

1. 4.48L **2.** 1.12L **3.** 0.224L
4.(a) 5.6L **(b)** 2.8L **5.** 2.8L **6.(a)** 1.12L **(b)** 6.9g
7. 3.36L **8.(a)** 2.24L **(b)** 6.72L **9.(a)** 336L **(b)** 1375g
10. 2.24L

SET 35

1. 0.2 **2.** 18g **3.** 14g
4.(a) 10.7g **(b)** 7.4g **5.** 7.1kg **6.(a)** 4 **(b)** 400g
7. 9.6g **8.** 15.6g **9.** 64.35g
10.(a) 3.175g **(b)** 9.375g

SET 36

1. 2.5L **2.** 0.42L **3.** 30mL
4. 10mL **5.(a)** 3000L **(b)** 1000L **6.** 50mL
7. 400mL *(200mL CO_2, 200mL H_2O)* **8.** 25ml O_2
9. 100mL *(60mL unused O_2, 40mL CO_2)* **10.** nil

SET 37

1.(a) 0.2 **(b)** 2.24L **2.(a)** 3.45g **(b)** 0.025 **(c)** 0.56L **3.(a)** 0.02 **(b)** 1.27g
4.(a) 0.5 **(b)** 0.75 **5.** 22.4g **6.(a)** 12.7g **(b)** 2.24L
7. 8g **8.** 0.635g **9.** 8.96L
10.(a) 25mL **(b)** 125mL **11.** 18g **12.(a)** 20.1g **(b)** 0.1 **(c)** 1.12L
13.(a) 33.6g **(b)** 13.44L **14.** 64mL **15.** 2.24L
16.(a) 3.8g **(b)** 0.08 **(c)** 1.792L **17.(a)** 140mL **(b)** 200mL *(80mL CO_2, 120mL H_2O)*
18. 0.112L **19.** 10.35g **20.** 15g